

$$- \textcircled{1} \Rightarrow u_x = g(y)$$

$$\Rightarrow u = g_1(y)x + f(y)$$

$$\textcircled{2} u = xy + u_1$$

$$u_{1,xy} = 0$$

$$u_1|_{x=0} = y \quad u_1|_{y=0} = x$$

$$\Rightarrow u_1 = x + y$$

$$u_1 = x + y + xy$$

(3) 特解 $\sin t$

$$u_{tt} = 4u_{xx}$$

$$u|_{t=0} = \sin x \quad u_t(0, x) = \sin x - 1$$

故特解

$$\sin t + \frac{1}{2}(\sin(x-2t) + \sin(x+2t)) \\ + \frac{1}{4} \int_{x-2t}^{x+2t} (\sin x - 1) dx$$

$$\sin t - t + \frac{1}{2}(\sin(x-2t) + \sin(x+2t)) \\ + \frac{1}{4}(\cos(x-2t) - \cos(x+2t))$$

$$\therefore \begin{cases} \Delta u = 0 & \text{与 } \varphi \text{ 无关} \\ u \end{cases}$$

$$\Rightarrow u = \sum_{n=0}^{+\infty} (A_n r^n + B_n r^{-n-1}) P_n(\cos \theta)$$

$$u_{r=1} = u_{r=2} = 2P_2(\cos \theta) + 1$$

$\Rightarrow$

$$\begin{cases} A_0 + B_0 = 1 \\ A_0 + \frac{1}{2}B_0 = 1 \end{cases}$$

$$\begin{cases} 4A_2 + \frac{1}{8}B_2 = 2 \\ A_2 + B_2 = 2 \end{cases}$$

$$\begin{cases} B_0 = 0 \\ A_0 = 1 \end{cases}$$

$$\begin{cases} \frac{31}{8}B_2 = 6 \\ A_2 = \frac{65}{48} \end{cases} \quad B_2 = \frac{31}{48}$$

原式

$$\Rightarrow \left(\frac{65}{48} r^2 + \frac{31}{48} r^{-3} \right) P_2(\cos \theta) + 1$$

$$\equiv \begin{cases} u_{te} = u_{rr} + \frac{1}{r} u_r \\ u|_{r=3} = 0 \\ u_{t=0} = 0 \quad u(t=0^-) = f(r-2) \end{cases}$$

$$\frac{T''}{T} = \frac{R''}{R} + \frac{R'}{rR} = -\lambda$$

$$\Rightarrow \text{解为 } \sum_{n=1}^{+\infty} J_0(\omega_n r) (A_n \cosh \omega_n t + B_n \sinh \omega_n t)$$

$$J_0(3\omega_n) = 0 \text{ 自由解, } u_{t=0} = 0$$

$$\Rightarrow \text{解为 } \sum_{n=1}^{+\infty} A_n \sinh \omega_n t J_0(\omega_n r)$$

$$\Rightarrow A_n = \frac{\int_0^3 r J_0(\omega_n r) f(r-2) dr}{\| \cdot \|}$$

$$A_n = \frac{J_0(2\omega_n)}{\frac{9}{2} J_1^2(3\omega_n) \omega_n} = \frac{4 \cdot J_0(2\omega_n)}{9 J_1^2(3\omega_n) \omega_n}$$

$$\sum_{n=1}^{+\infty} \frac{4 J_0(2\omega_n)}{9 J_1^2(3\omega_n) \omega_n} \sinh \omega_n t J_0(\omega_n r)$$

其中 ω_n 为 $J_0(3\omega_n) = 0$ 的第 n 个解

四. $\frac{d\bar{U}}{dt} = (-2\lambda^2 - 4^2 + 1)\bar{U}$

$$\bar{U}_{t=0} = (2\pi)^2 \frac{1}{2\sqrt{\pi}} \exp(-\frac{\lambda^2}{4}) \frac{1}{2\sqrt{\pi}} \exp(-\frac{4^2}{4})$$

$$= \pi \exp(-\frac{\lambda^2}{4} - \frac{4^2}{4})$$

$$\bar{U} = \pi \exp\left[-(2t + \frac{1}{4})\lambda^2 - (t + \frac{1}{4})4^2 + t\right]$$

$$\Rightarrow \bar{U} = \pi \frac{1}{\sqrt{8t+1} \sqrt{4t+1}} \exp\left(-\frac{\lambda^2}{8t+1} - \frac{4^2}{4t+1} + t\right)$$

① $\lambda - 1 > 0$ 时

五. X' 的解 $e^x (A \cos vx + B \sin vx) \Rightarrow$ 解 $e^x \sin(n\pi)x$

①

$$\lambda = (n\pi)^2 + 1$$

② $\lambda - 1 = 0$ 时 解 $Ax + B \Rightarrow A = B = 0$ (舍)

③ $\lambda - 1 < 0$ 时 $A_1 e^{v_1 x} + B_1 e^{v_2 x}$ (代 λ $A_1 = B_1 = 0$)
 $v_1 \neq v_2$

$$\textcircled{2} \quad \frac{T'}{T} = \frac{X''}{X} + 2\frac{X'}{X} = -\lambda$$

$$\Rightarrow \text{解为 } \sum_{n=1}^{+\infty} e^x \sin(n\pi x) e^{-(n\pi)^2 t} \cdot A_n$$

$$u(0, x) = f(0, \frac{1}{2})$$

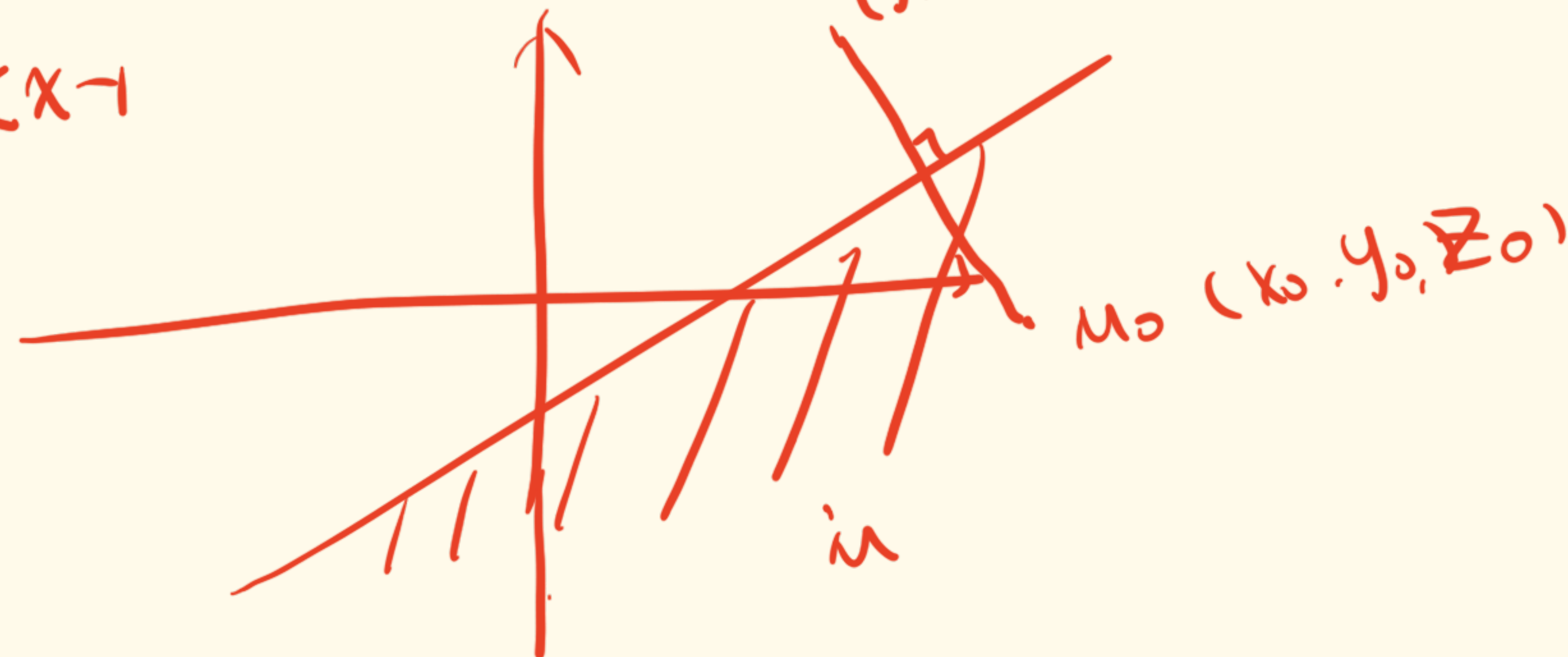
$$\Rightarrow A_n = 2 \int_0^1 e^{2x} \sin n\pi x \cdot f(x - \frac{1}{2}) dx$$

$$= A_n = 2 \cdot e \sin \frac{n\pi}{2}$$

$$\sum_{n=1}^{+\infty} \frac{2e \sin \frac{n\pi}{2}}{(y_0+1, x_0-1) z_0} e^{-x} \sin n\pi x e^{-(n\pi)^2 t}$$

六.

$y < x-1$



$$\textcircled{1} \text{ 解为 } \frac{G(M, M_0)}{4\pi} \frac{1}{\sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2}} - \frac{1}{\sqrt{(x-y_0-1)^2 + (y-x_0+1)^2 + (z-z_0)^2}}$$

$$\textcircled{2} \quad x=y+1 \text{ 且外法向量 } (-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0)$$

故而解为

$$\int \varphi(x_0, x_0+1) \left(\frac{\partial G}{\partial x_0}, \frac{\partial G}{\partial y_0}, \frac{\partial G}{\partial z_0} \right) \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0 \right) \sqrt{2} dx_0 dz_0$$

$$\frac{\partial G}{\partial y_0} - \frac{\partial G}{\partial x_0} = \frac{-1}{2} \left(\frac{x-1-y}{\sqrt{(x-x_0)^2 + (y-x_0+1)^2 + (z-z_0)^2}^3} - \frac{y+1-x}{\sqrt{(x-x_0)^2 + (y-x_0+1)^2 + (z-z_0)^2}^3} \right)$$

$$= \frac{y+1-x}{\left(\sqrt{(x-x_0)^2 + (y-x_0+1)^2 + (z-z_0)^2} \right)^3}$$

$$\Rightarrow \text{原式} = \frac{1}{4\pi} \int \varphi(x_0, x_0+1) \frac{(x-1-y)}{\left(\sqrt{(x-x_0)^2 + (y-x_0+1)^2 + (z-z_0)^2} \right)^3} dx_0 dz_0$$

$$\text{代 } \lambda z = \frac{z}{2}$$

$$= \frac{1}{4\pi} \int \rho(x_0, x_0-1) \frac{(-x+1+y)}{\left(\sqrt{x-x_0+1} \left(y-x_0+1 \right)^2 + \left(\frac{z}{2} - z_0 \right)^2 \right)^{3/2}} dx_0 dz_0$$